

Importing Required Libraries and Modules

In this phase, all the required libraries and modules are imported so that the application development and machine learning implementation can be performed smoothly without errors. Importing the necessary packages at the beginning of the project ensures that the required functions, classes, and utilities are available throughout the development process.

Common Libraries Used

Library / Module	Purpose
NumPy	Numerical computations and array operations
Pandas	Data loading, cleaning, and manipulation
Matplotlib	Data visualization and plotting
Seaborn	Statistical data visualization
Scikit-learn	Machine learning model development and evaluation
Flask	Web application framework
Joblib	Saving and loading trained machine learning models
OS	File and directory operations
Pickle (Optional)	Model serialization

Benefits

- Ensures all dependencies are available before execution.
- Prevents runtime import errors.
- Simplifies data preprocessing, model training, prediction, and visualization.
- Improves code organization and maintainability.